Chapter 23: Wave Properties of Light

In this chapter we will study the following:

- Index of Refraction
- Spectrum of Light
- Huygens' Principle
- Reflection, Refraction and Total Internal Reflection of light
- Young's Double Slit (interference of Light)
- Diffraction Grating
- Mercury Spectrum
- Diffraction by a Narrow Slit
- Diffraction by a Circular Aperture

23.1 The index of Refraction:

The index of refraction n of the medium is Defined as $n = c/v$, where
C = speed of light in vacuum
V = speed of light in the medium

The frequency f of the light is related to its speed by $V = \lambda f$, Where λ is the wavelength of the light measured in meters
units of f is s^{-1} = Hz
Speed of light = $C = 3 \times 10^8 m/s$

Index of refraction for some materials:

TABLE 23.1

Indices of refraction of representative materials for yellow sodium light (λ = 589 nanometres) We use the approximate values n = 1 for air and n = 4/3 for water.

Material	Index
Air	1.00029
Carbon dioxide	1.00045
Water	1.333
Ethyl alcohol	1.362
Benzene	1.501
Carbon disulfide	1.628
Glass, light crown	1.517
Glass, heavy flint	1.647
Fluorite	1.434
Diamond	2.417

Spectrum of Light

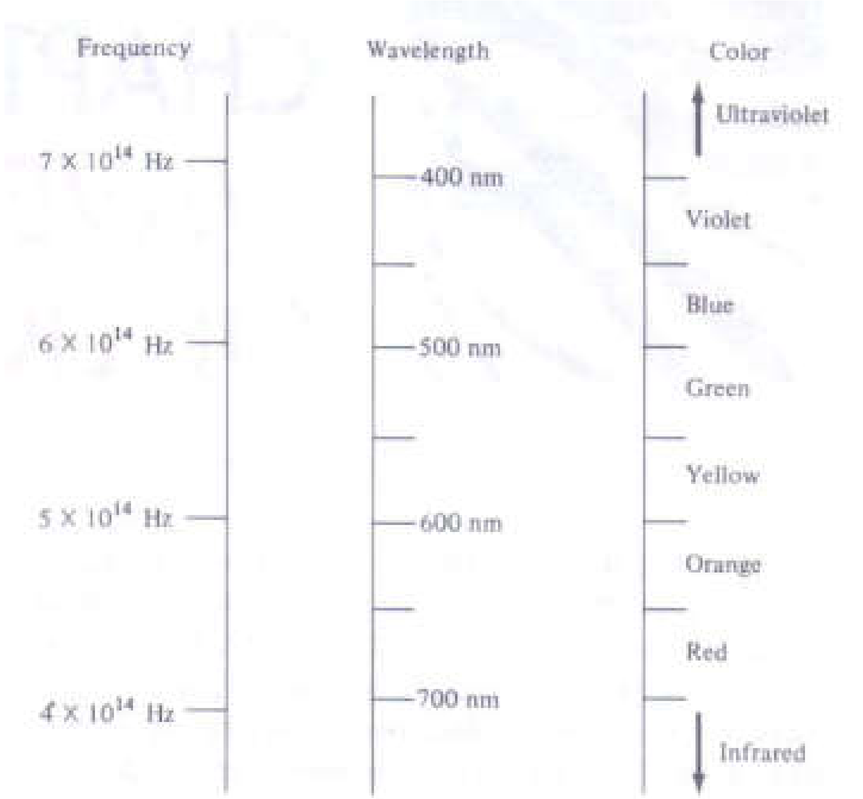
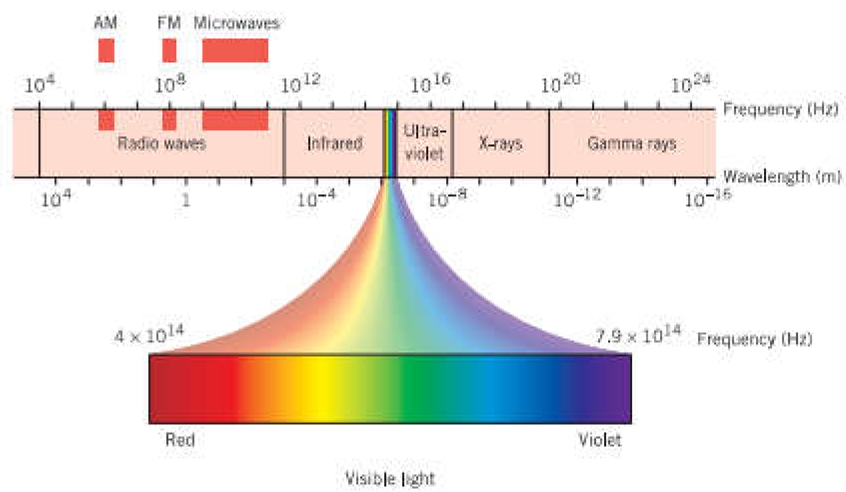


Figure 23.1. The spectrum of visible light. Color ranges are approximate.

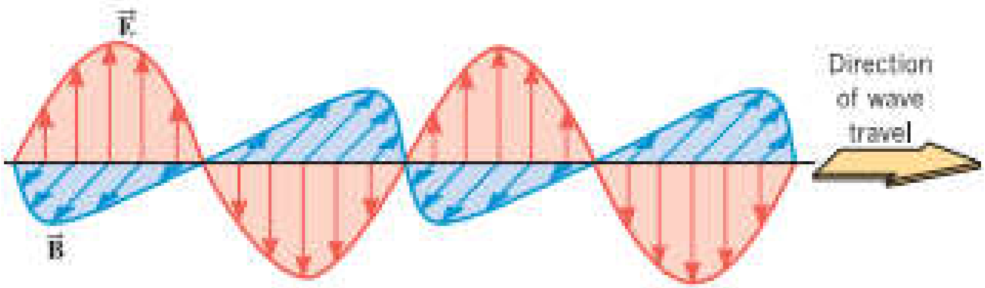


If a beam of light goes from a medium with refractive index n_1 to a medium with index n_2

$$\text{then} \quad f\lambda_1 = \frac{c}{n_1} \quad \text{and} \quad f\lambda_2 = \frac{c}{n_2}$$

$$\text{divide, this gives} \Rightarrow \frac{f\lambda_1}{f\lambda_2} = \frac{c/n_1}{c/n_2} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1}$$

- **This means that the wavelength is smaller in the optically denser medium**



Light is an electromagnetic waves

EXAMPLE 23.1 Green light with a wavelength of 5×10^{-7}m in vacuum enters a glass plate with refractive index 1.5 (a) What is the velocity of light in glass? (b) What is the wavelength of light in the glass?

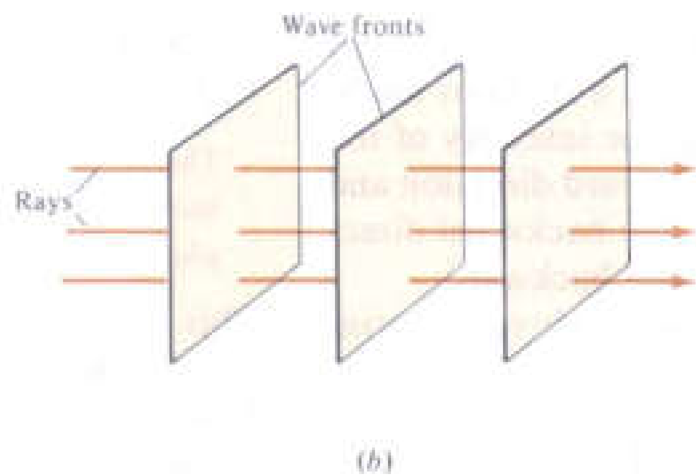
(a) $$V = \frac{c}{n} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 m/s$$

(b) $$In\ Vacuum\ V = c = 3 \times 10^8 m/s$$

$$So,\ \lambda_2 = \lambda_1 \times \frac{n_1}{n_2} = 5 \times 10^{-7} \times \frac{1}{1.5} = 3.3 \times 10^{-7} m$$

23.2 Huygens' Principle

Huygens' principle, also called Huygens-Fresnel principle, a statement that all points of a wave front of sound in a transmitting medium or of light in a vacuum or transparent medium may be regarded as new sources of wavelets that expand in every direction at a rate depending on their velocities



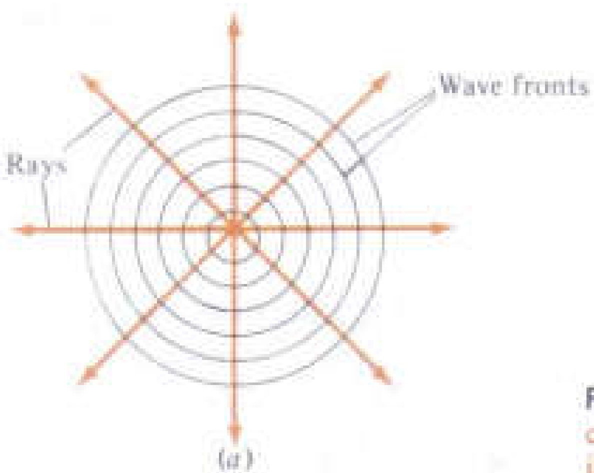
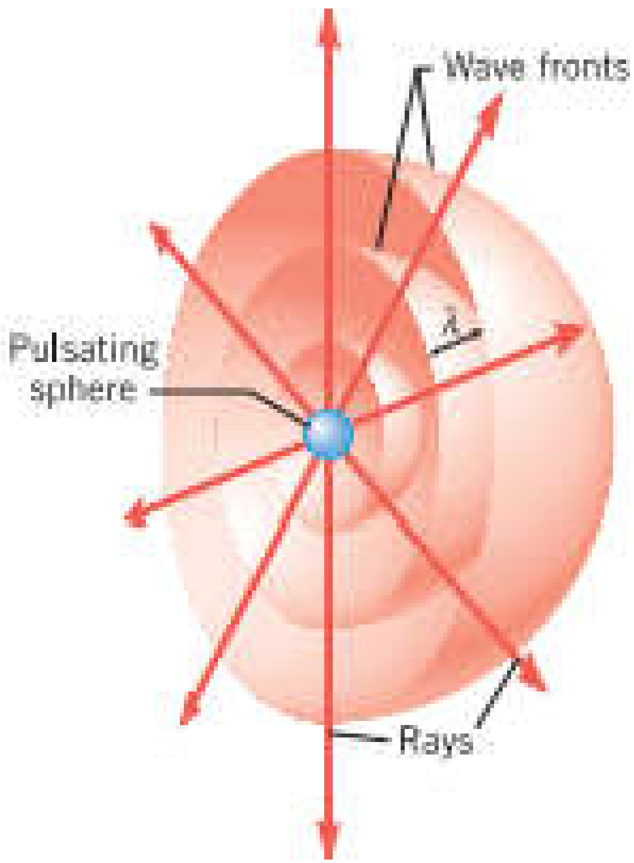


Figure 23.2. (a) The wave fronts corresponding to successive maxima in a spherical wave. (b) Wave fronts for a plane wave.



A hemispherical view of a sound wave emitted by a pulsating sphere. The wave fronts are drawn through the condensations of the wave, so the distance between two successive wave fronts is the wavelength λ. The rays are perpendicular to the wave fronts and point in the direction of the velocity of the wave.

23.2 Reflection of Light

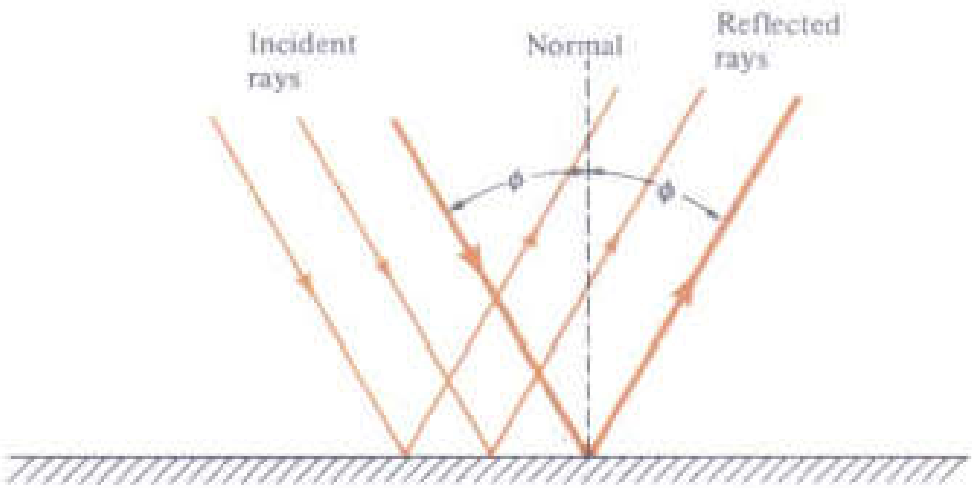


Figure 23.5. The reflected light rays are in the same plane as the incident rays and the normal and make the same angle with the normal.

The reflected light rays are in the same plane as the incident rays and the normal, and make the same angle with the normal

θ(incident) = θ(reflected)

In Fig. $\theta = \Phi$

The intensity I_r of the reflected wave is related to the intensity I_o of incident wave by the following relation:

$$R = \frac{I_r}{I_o} = \left(\frac{n_2 - n_1}{n_2 + n_1}\right)^2$$

R is the reflectance

n_1 is the refractive index of reflected medium and n_2 is the refractive index of the refracted medium

The above relation is correct for normal incidence

The transmittance T is the ratio of incident intensity I_o to the refracted intensity I_t and is equal

$$T = \frac{I_t}{I_o} = 1 - R$$

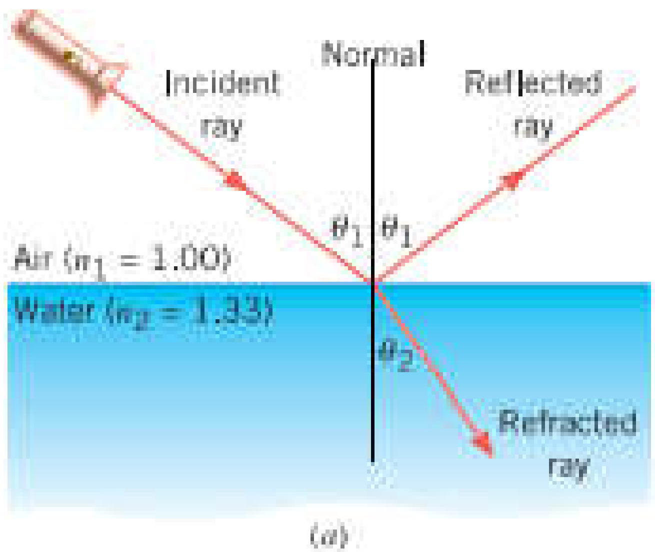
EXAMPLE 23.2 A lens is made of glass with n = 1.5. What Fraction of the light is reflected at normal incidence?

Solution: with n_1 =1 and n_2 = 1.5

$$\frac{I_r}{I_o} = \left(\frac{n_2 - n_1}{n_2 + n_1}\right)^2 = \left(\frac{1.5 - 1}{1.5 + 1}\right)^2 = 0.04$$

It means that 4% of light is reflected and 96% of light is transmitted T = 0.96

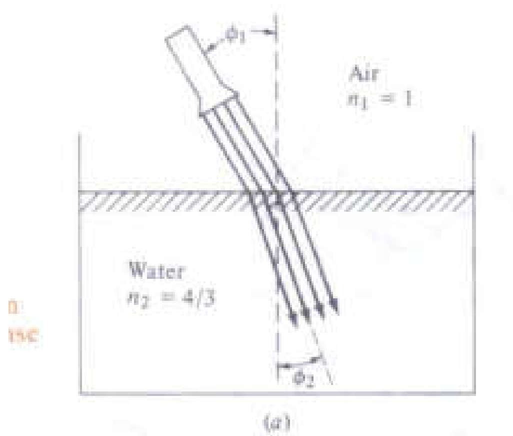
23.4 Refraction of light



(a)

Light incident from air into water, part of light is reflected at the interference and the reminder is refracted

The angle of incidence related to the angle of refracted by Snell's Law:



(a)

$$n_1 \sin \Phi_1 = n_2 \sin \Phi_2$$

Φ_1 is the incident angle

Φ_2 is the refracted angle

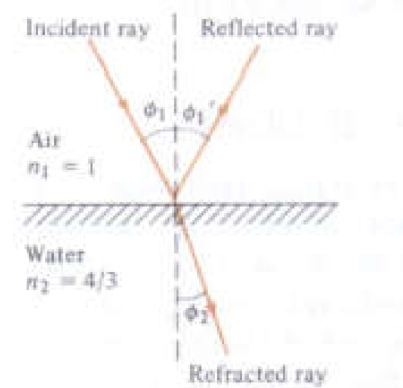
EXAMPLE 23.3 A beam of light incident from air on water at angle of 30°. Part of the light is reflected and part is refracted (Fig. below). Find the angles of the two beams.

Solution:

- The angle of incidence is equal to the angle of reflection

$$\Phi' = \Phi_1 = 30^o$$

- Apply Snell's law to get the angle of refraction



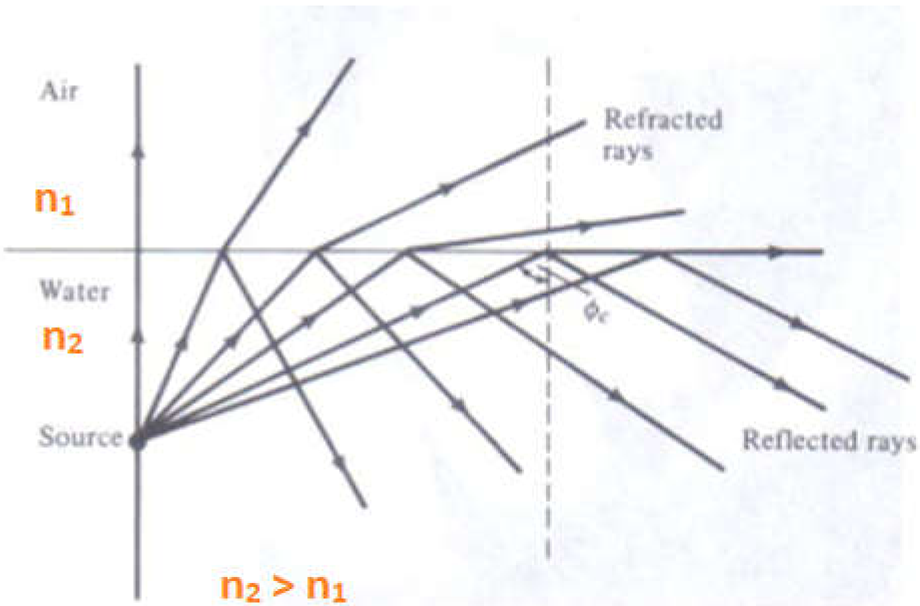
$$n_1 sin\Phi_1 = n_2 sin\Phi_2$$

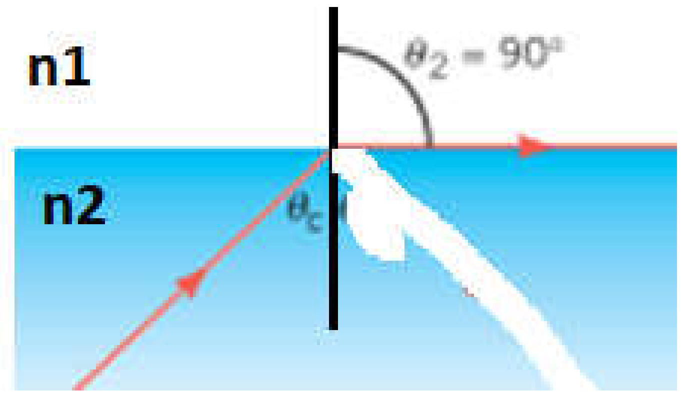
$$n_1 \sin\Phi_1 = n_2 \sin\Phi_2$$

$$\Rightarrow \sin\Phi_2 = \frac{n_1}{n_2}\sin\Phi_1 = \frac{1}{4/3}\sin 30^o = 0.375$$

$$\Phi_2 = \sin^{-1}(0.375) = 22^o$$

23.5 Total Internal Reflection





Light incident from medium of $n_2 > n_1$, at some incident angle the light suffers from total reflection. Greater then some critical angle θ_c light does not pass through medium n_1

Conditions for total reflection:

(1) The incident angle greater than some critical angle θ_c

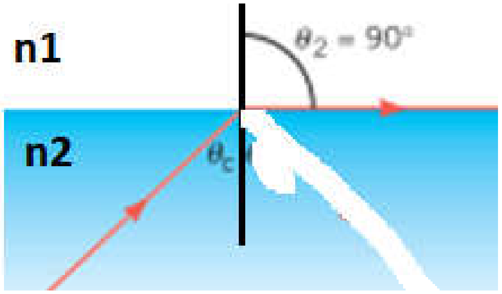
(2) The refractive index of incident medium n_2 is greater than the index n_1 (see Figs.)

The critical θ_c can be determined using Snell's law. At θ_c the refracted angle is at 90°.

$n_1 \sin\theta_1 = n_2 \sin\theta_2$

n_1 is the refracted medium and n_2 is the incident medium(see previous slide)

$\theta_1 = 90^o$, $\theta_2 = \theta_c$

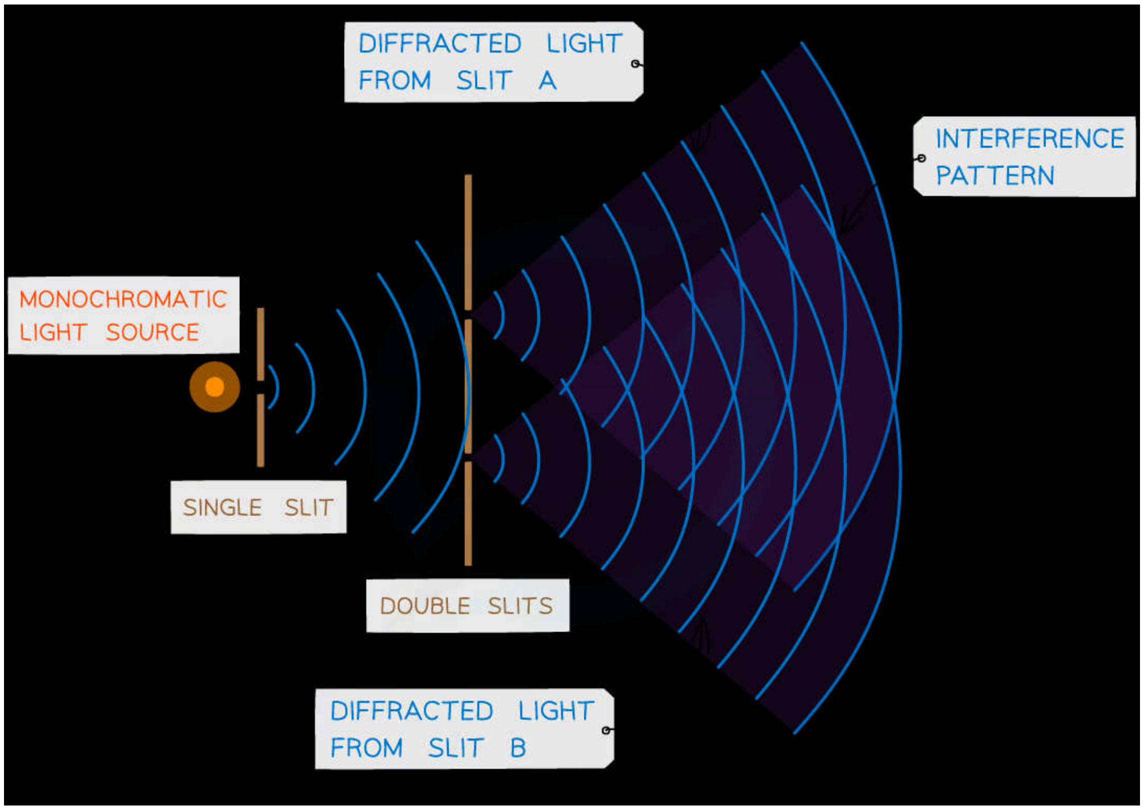


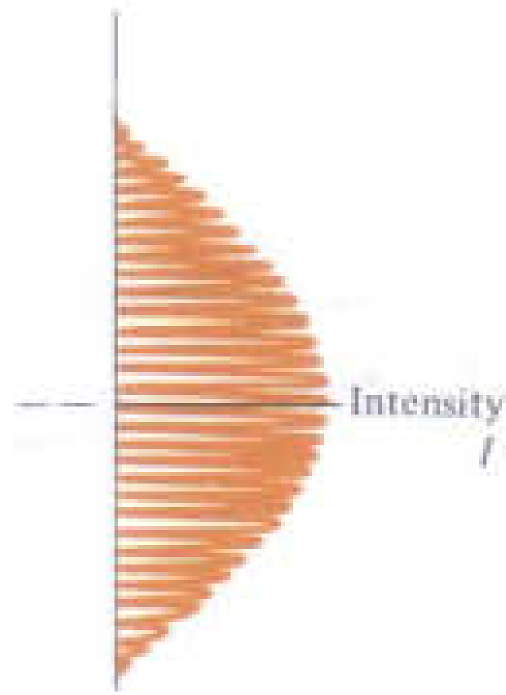
$n_1 \sin 90^o = n_2 \sin\theta_c \Rightarrow \sin\theta_c = n_1/n_2$ NOTE: $n_2 > n_1$

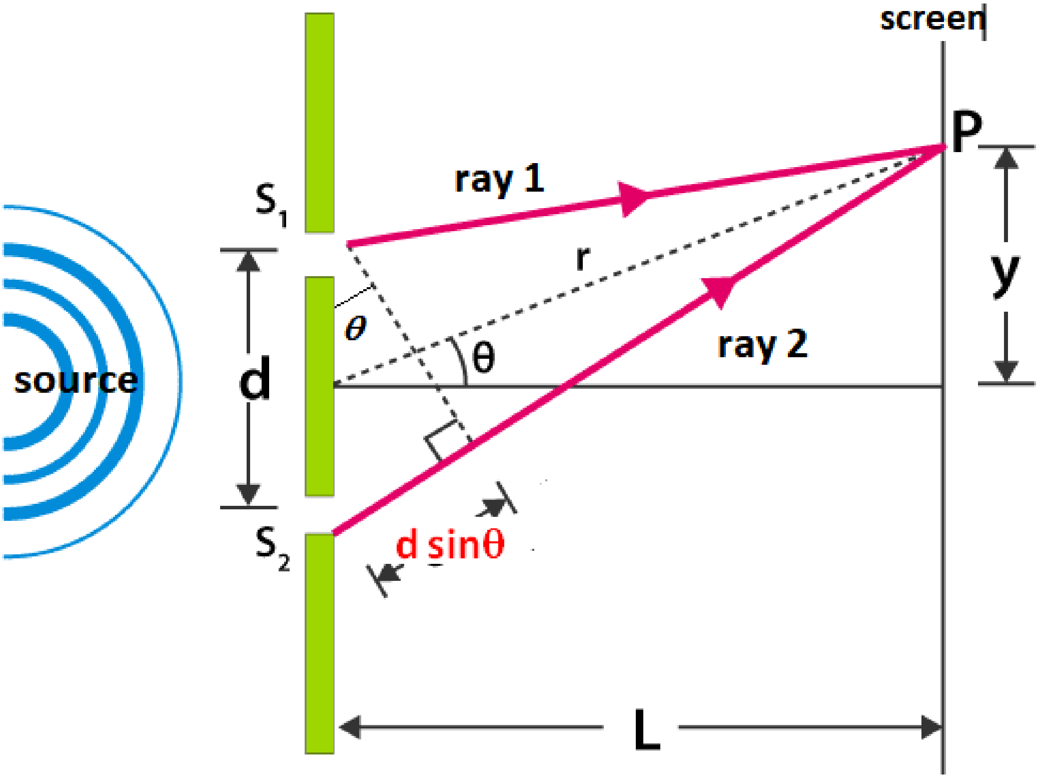
Example: What is the critical angle θ_c for light from glass (n_2 = 1.5) into air (n_1 = 1)?

$\sin\theta_c = n_1/n_2 = 1/1.5 = 0.667 \Rightarrow \theta c = 42^o$

23.6 Young's Double Slit Interference Experiment







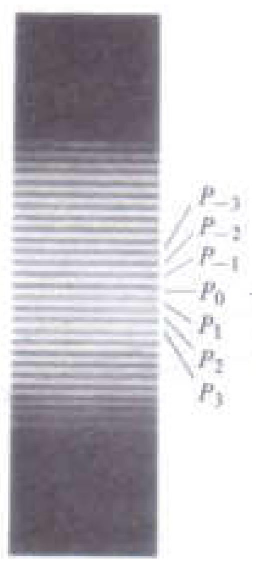
When the path difference between ray1 and ray2 is equal multiple of the wavelength λ of the light (i.e., λ, 2λ, 3λ, ..$m\lambda$, m integer) constructive interference occurs at the screen.

Path difference = $d \sin\theta$

$d \sin\theta = m \lambda$, $m = 0, \pm1, \pm2, \ldots$

m= order of the fringe

NOTE: θ is small angle



Typical fringes seen on the screen, bright fringes occur at maxima

For maximum intensity (white fringes) of light on the screen, we have $d \sin\theta = m\lambda$, $m = 0, \pm1, 2\pm, ,\ldots$ maxima

NOTE: Dark fringes occur when the path difference is $\lambda/2$, $3\lambda/2$, …

EXAMPLE 23.6 Find the angle of the m =3 fringe if two slits 0.4mm apart are illuminated by yellow light of wavelength 600nm.

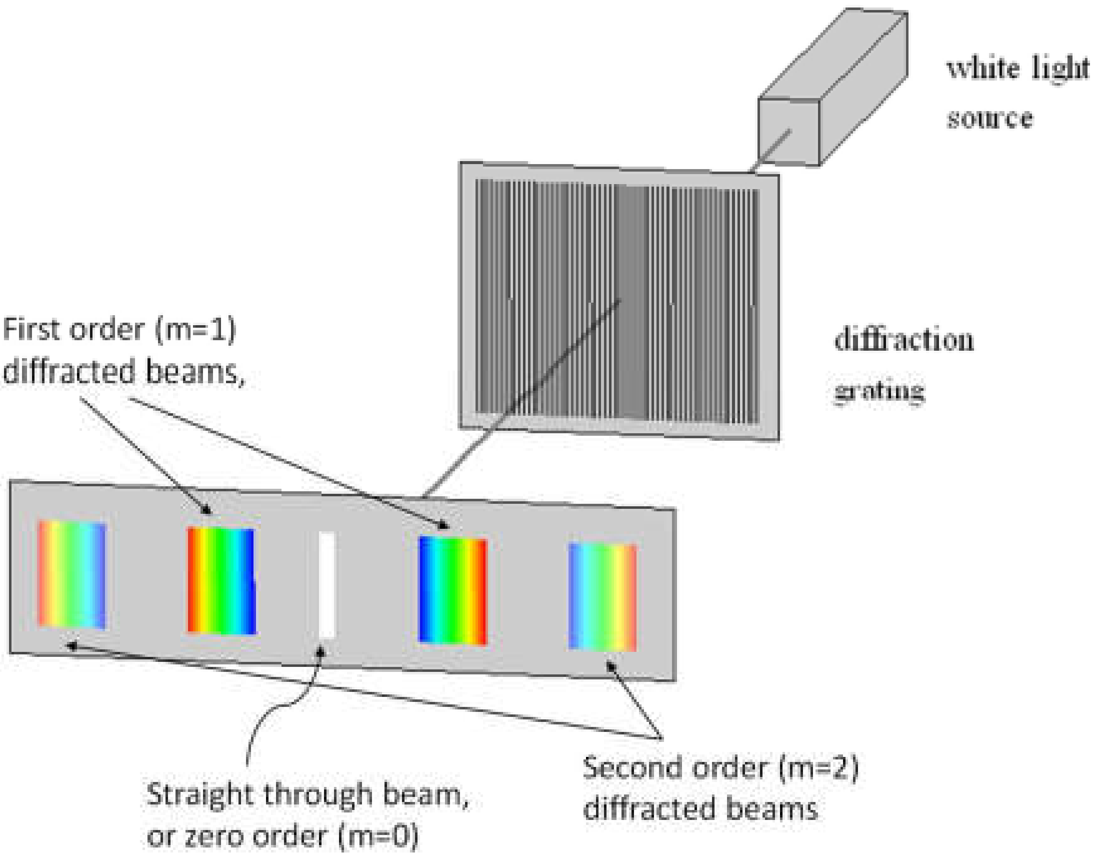
Solution: NOTE 1mm = 10^{-3}m and 1nm =10^{-9}m

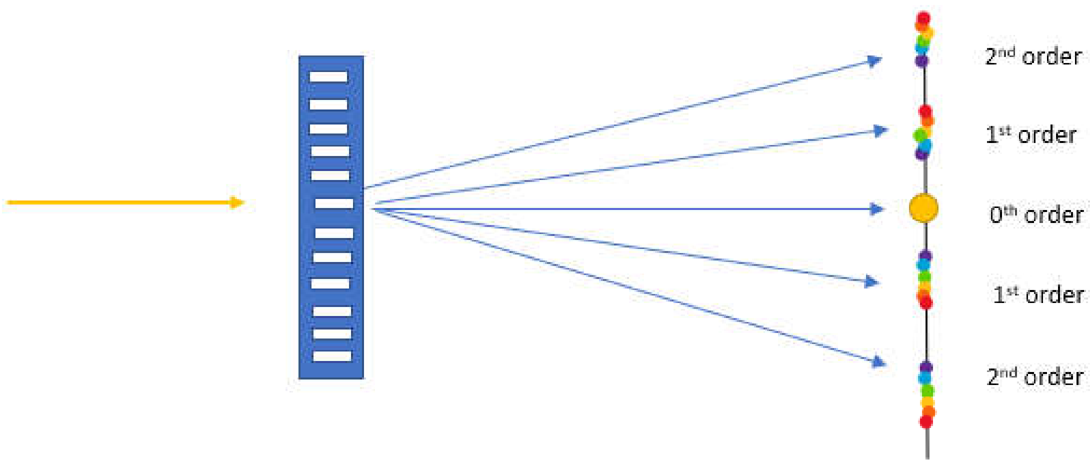
$$d sin\theta = m\lambda \rightarrow sin\theta = \frac{m\lambda}{d} = \frac{3\times(600\times10^{-9})}{4\times10^{-4}} = 4.5\times10^{-3}$$

Since θ small angle sinθ = θ = 4.5×10^{-3} rad×(180°/3.14 rad)
= 0.258°

NOTE: θ is small angle means that sinθ = tanθ = y/L (see previous slide)
d sinθ =m λ → d(y/L) = m λ → $y = m\,\lambda\,(L/d)$. Y is measured from central Bright fringe i.e., m = 0. (See Fig. in previous slide)

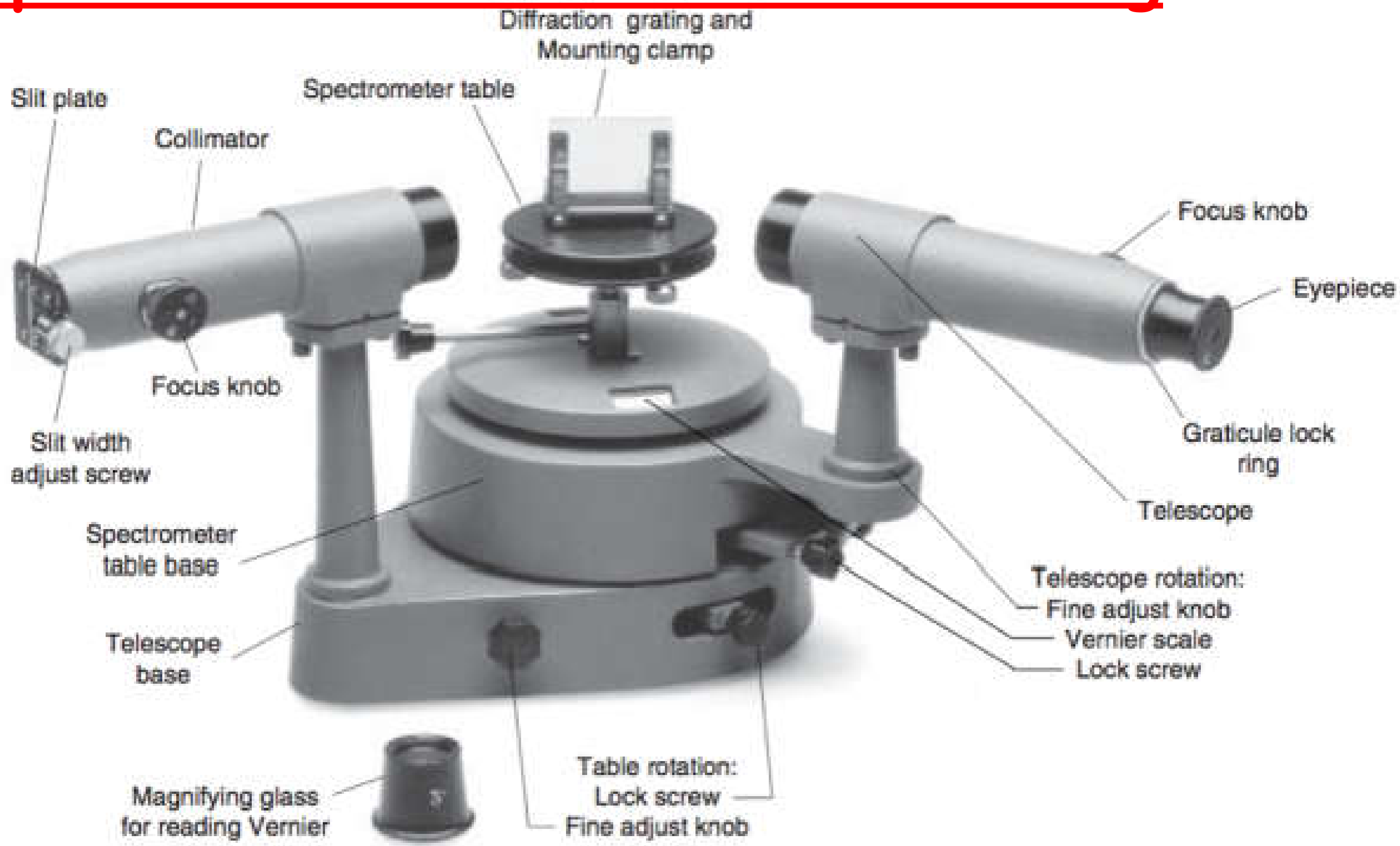
23.8 The Diffraction Grating



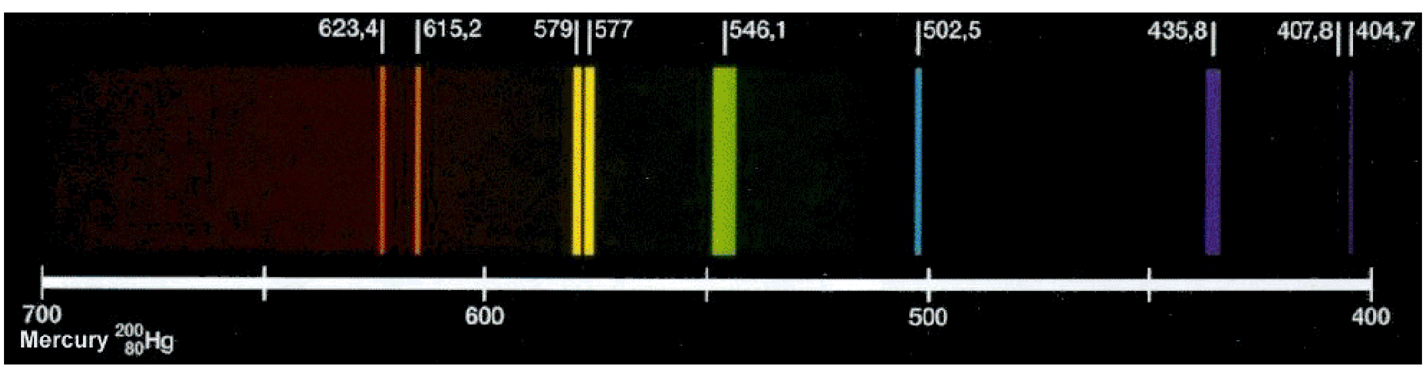


When white light is incident on a parallel grating plate with several or even hundreds of evenly-spaced identical slits, it is diffracted creating spherical waves around the openings that interfere with one another. This creates an interference pattern, where each wave interacts with another. This further creates a pattern of maximums and minimums

Apparatus used in Diffraction Grating



Mercury Spectrum



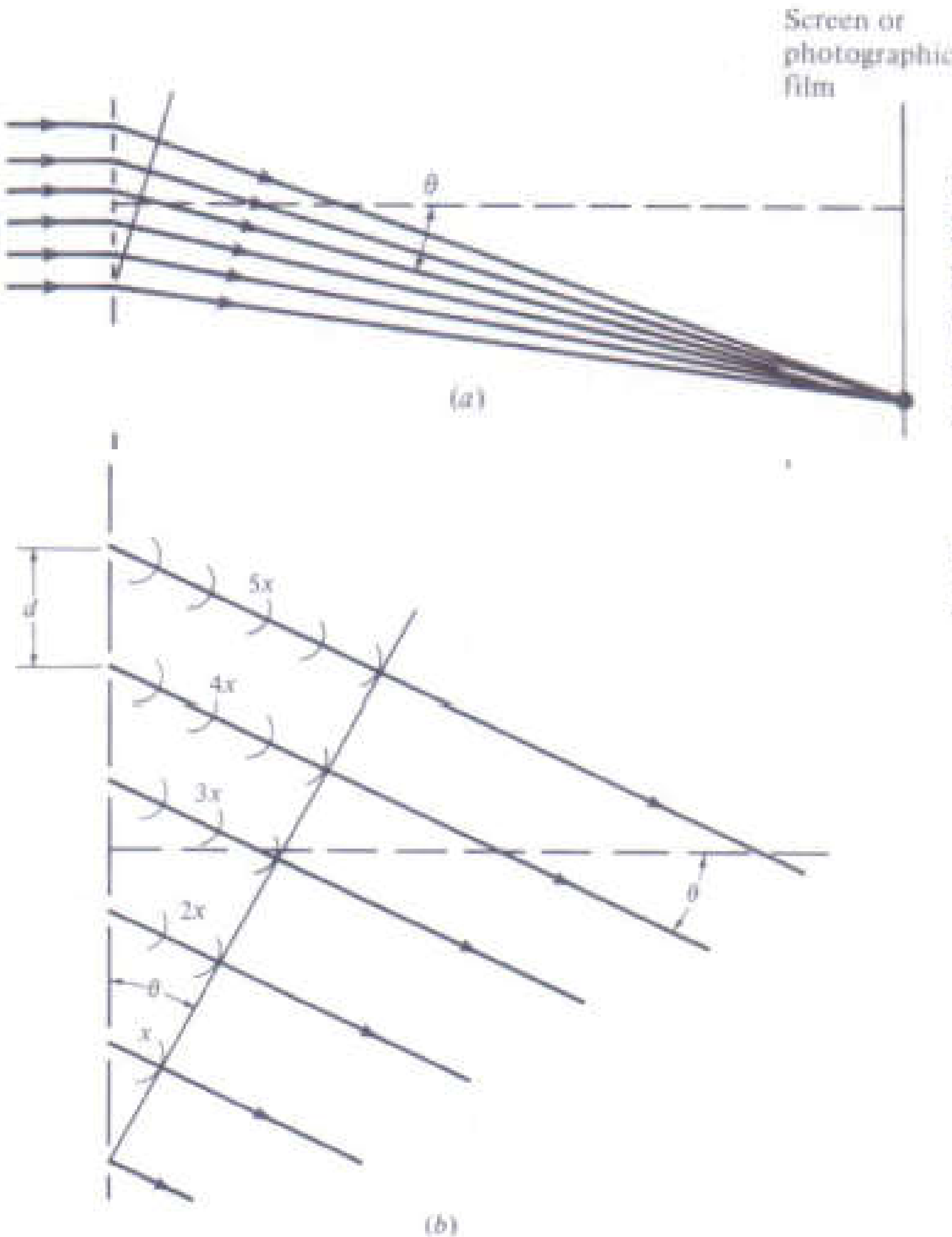


Figure 23.19. (a) Light rays reaching a point P on a screen from six closely spaced slits. (Light rays reaching other points are not shown.) (b) Enlarged view of the region near the slits.

The principle of the diffraction grating is illustrated by considering the case of six slits (Fig. 23.19). If the difference $x = d \sin \theta$ of the distances from a point on the screen to any two adjacent slits is exactly an integral multiple of a wavelength, all the waves reaching the screen are in phase. Thus a maximum in the intensity occurs when

$$d \sin \theta = m\lambda, \qquad m = 0, \pm 1, \pm 2, \ldots \qquad (23.8)$$

m is called the *order* of the maximum or spectrum produced by the grating.

The angles at which the maximum intensity points occur are known as fringes, and can be calculated using the grading equation below.

$d \sin\theta = m\lambda$

where d is the spacing between the slits in meters, θ is the separation angle between the order of maximum in degrees, n is the order of maximum, and ʎ is the wavelength of the source in meters.

EXAMPLE 23.8 A grating has 4000 lines per centimeter. At what angles are maxima formed if it is illuminated With yellow light at 600nm

Solution: The slit spacing is

$$d = \frac{1}{4000}cm = 2.5\times10^{-4}cm = 2.5\times10^{-6}m$$

$$d\sin\theta = m\lambda \Rightarrow \sin\theta = \frac{m\lambda}{d}$$

$$\sin\theta = m\times\frac{600\times10^{-9}m}{2.5\times10^{-6}m} = 0.24m$$

Diffraction by a Narrow Slit

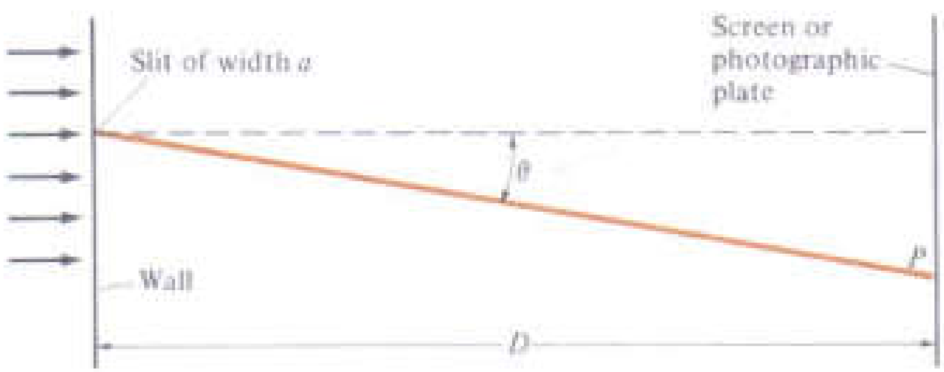


Figure 23.24. Geometry of the single-slit different experiment. The pattern seen on the screen is shown in Fig. 23.25.

$$a \sin\theta = m\lambda, \quad m = \pm 1, \ \pm 2, \ldots$$

for diffraction minima

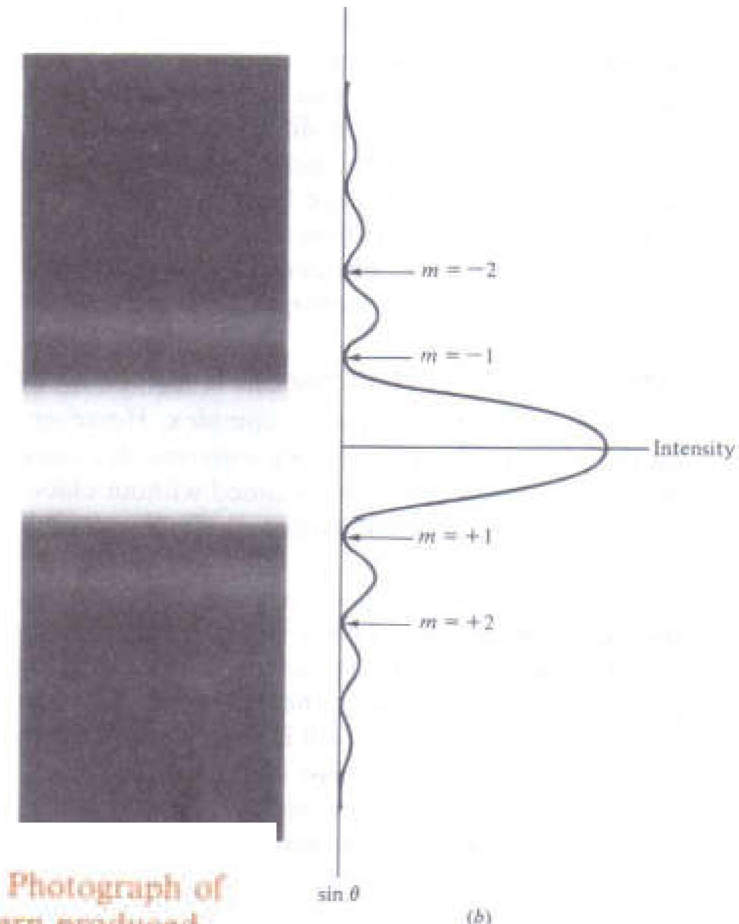
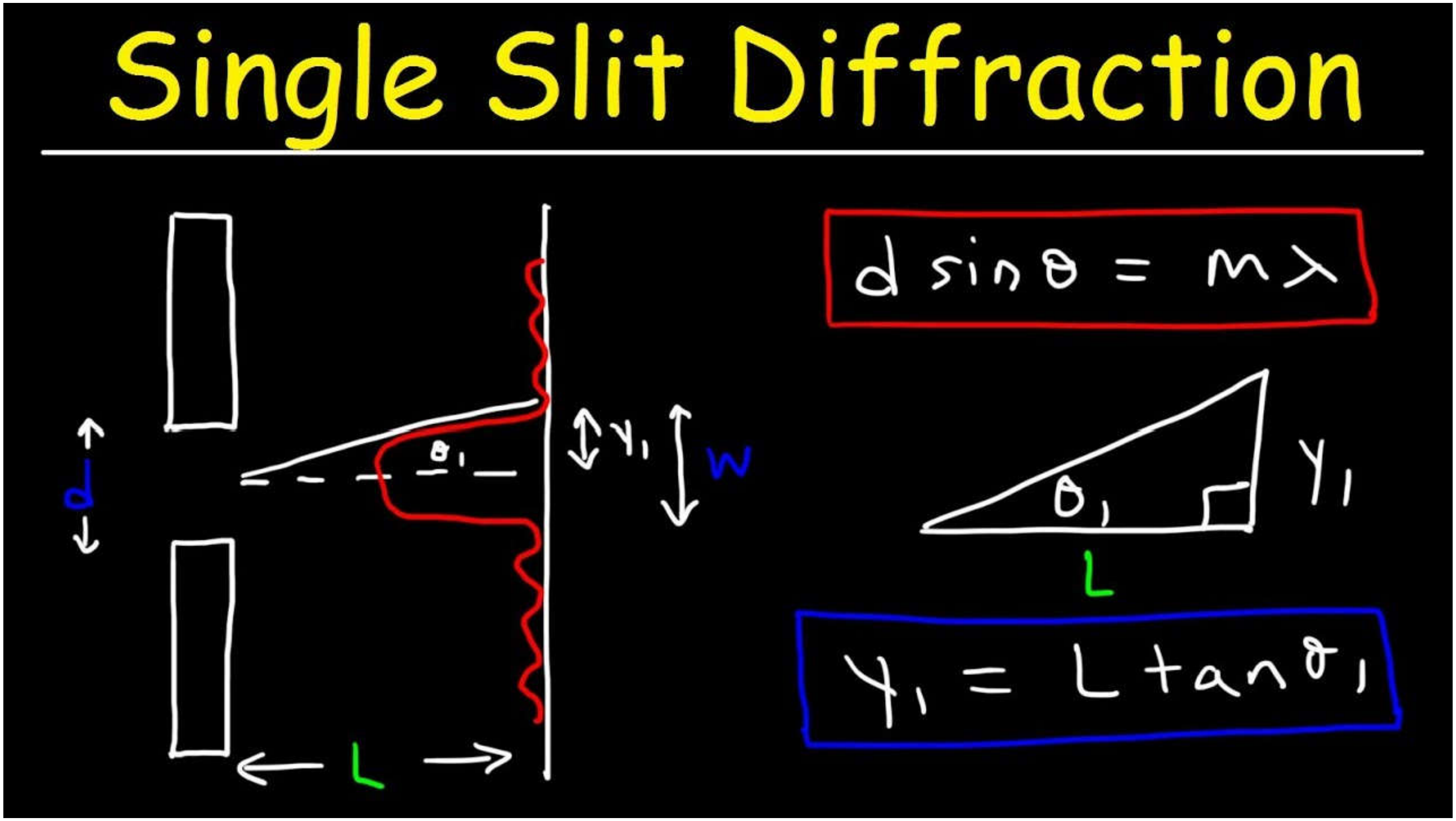
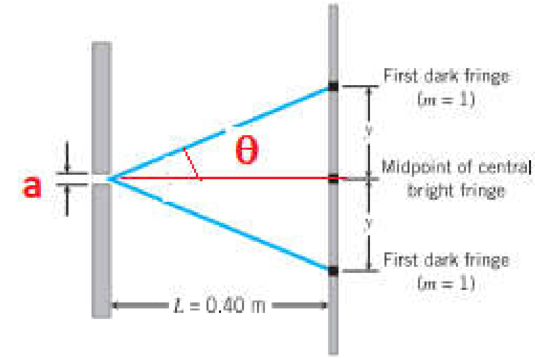


Figure 23.25. (a) Photograph of the diffraction pattern produced by a single slit illuminated by monochromatic light as in Fig. 23.24. (b) Intensity graph for this diffraction pattern. (From Sears, Zemansky, and Young, 5th ed., 1976, Addison-Wesley, Reading, Mass.)


Single Slit Diffraction
d
θ₁
y₁
W
L
d sinθ = mλ
θ₁
y₁
L
y₁ = L tanθ₁

Example: Light passes through a single slit and shines on a flat screen that is located **L** = 0.4m a way from the screen (see Fig.). The wavelength of the light in a vacuum is λ = 410nm. The distance between the midpoint of the central bright fringe and the first dark fringe y. Determine the **2y** of central bright fringe when the width of the slit is (a) **a** = 5×10^{-6}m and (b) **a** = 2.5×10^{-6}m.



The distance 2y is the width of the central bright fringe

a sinθ =λ , m =1

$$\theta = sin^{-1}\frac{\lambda}{a} = sin^{-1}\frac{410\times10^{-9}}{5\times10^{-6}} = 4.7^0$$

(a) $2y = 2L\tan\theta = 2\times0.4\tan4.7 = 0.066m$

(b) $$\theta = sin^{-1}\frac{\lambda}{a} = sin^{-1}\frac{410\times10^{-9}}{2.5\times10^{-6}} = 9.3^o$$

$$2y = 2L\tan\theta = 2\times0.4\tan9.3 = 0.13m$$

Diffraction by a Circular Aperture

Analysis show that the first minimum for a circle with diameter d occurs when:

$$\sin\theta = 1.22\frac{\lambda}{d}$$

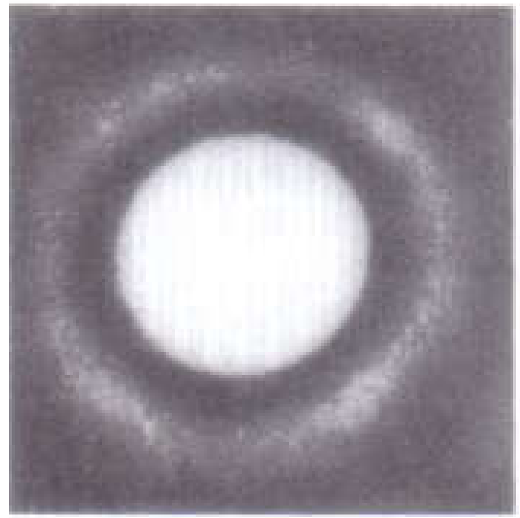
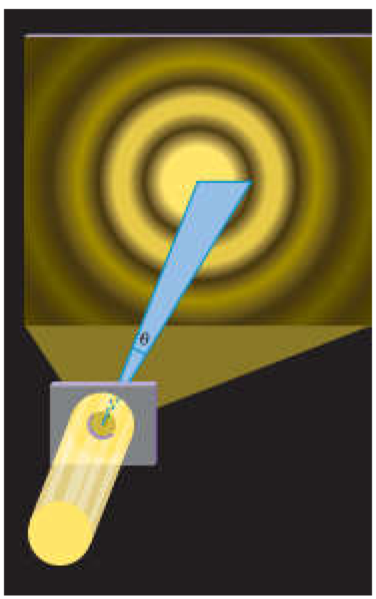
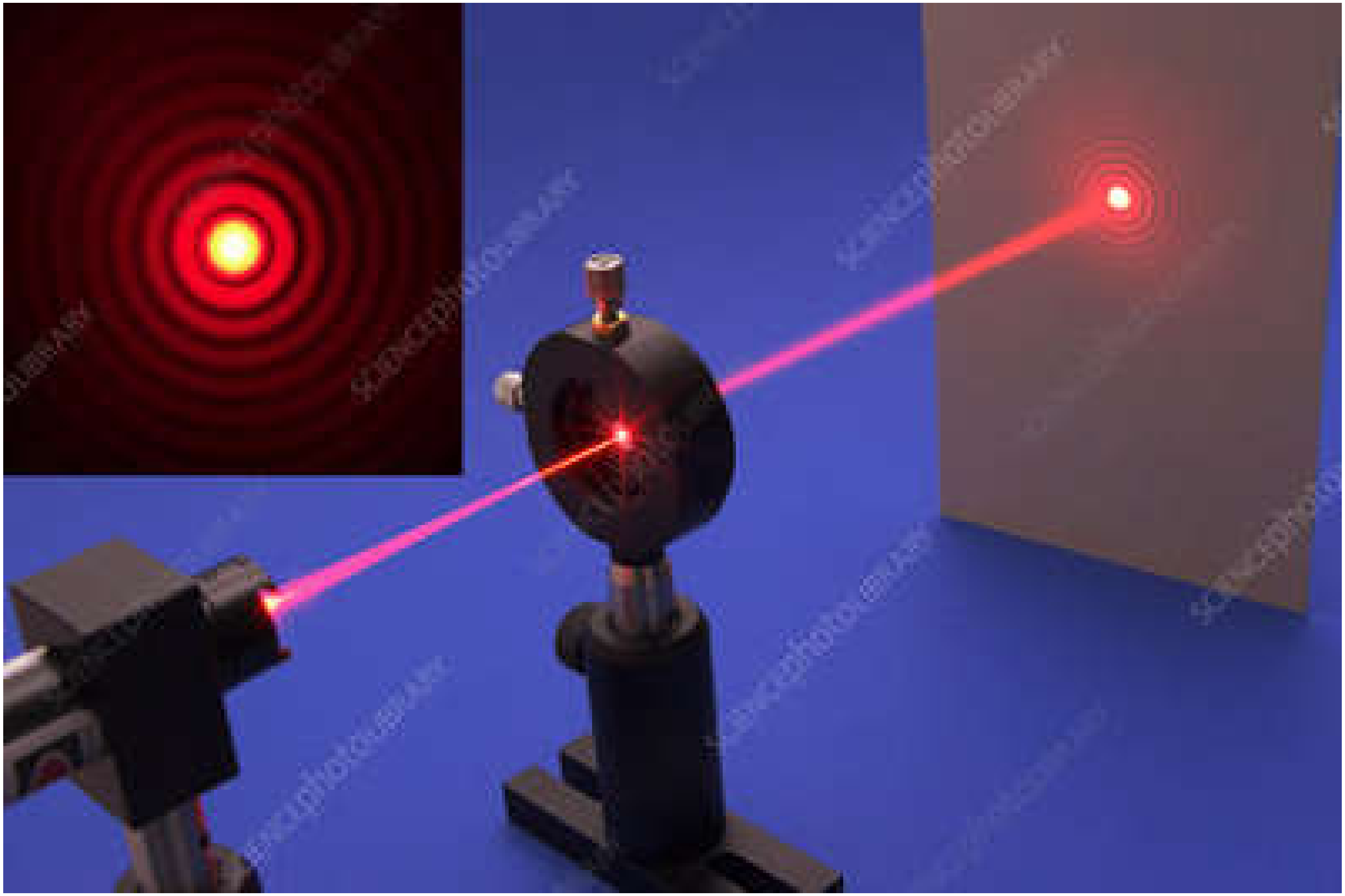


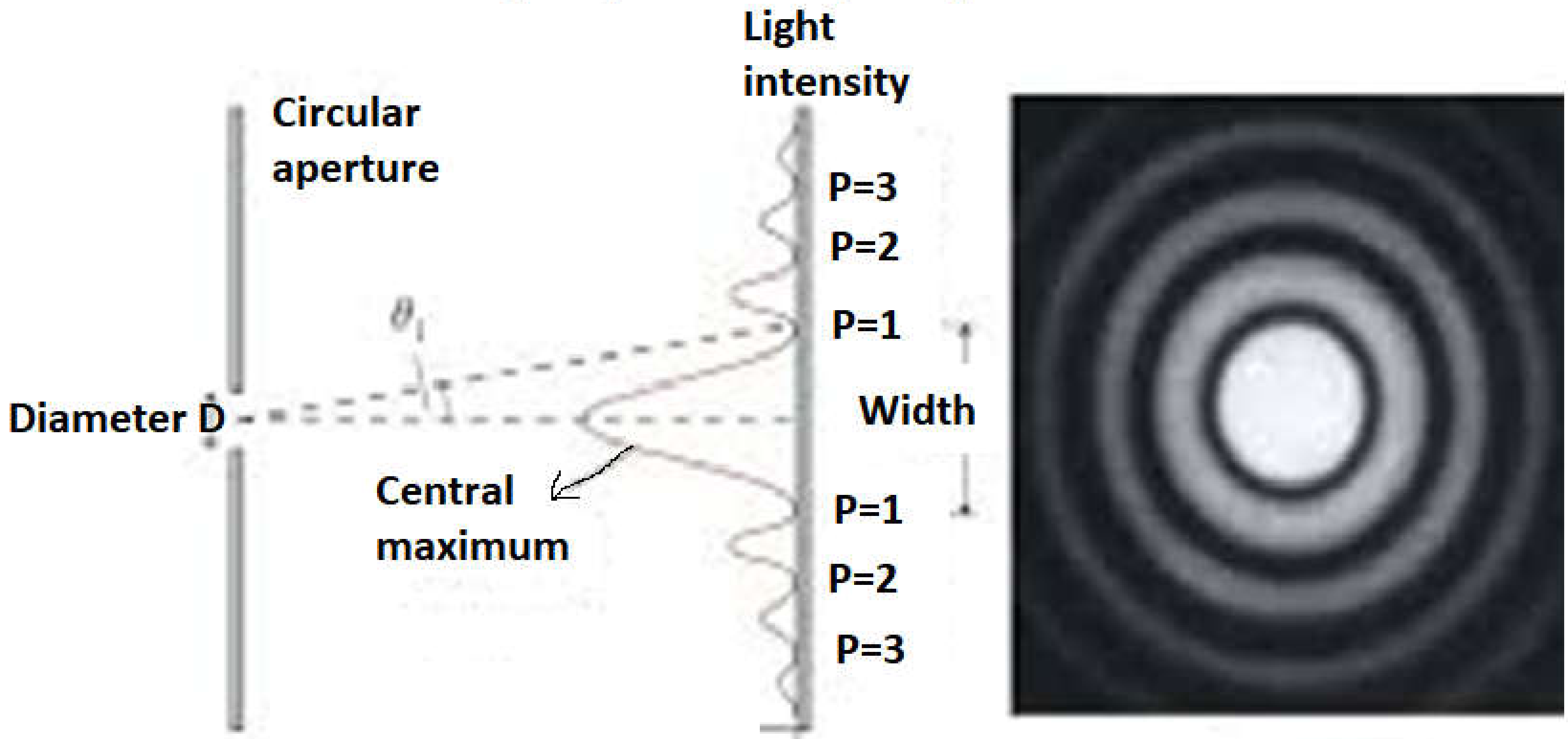
Figure 23.28. The image of a star formed on a photographic plate by a telescope. A central diffraction maximum and a circular secondary maximum are visible. Additional maxima also occur, but their intensities are too small to be seen here. (From Halliday and Resnick, *Physics*, Part II, 3rd ed. Copyright © 1978, John Wiley & Sons, New York.)



When light passes through a small circular opening, a circular diffraction pattern is formed on a screen. The angle θ locates the first dark fringe relative to the central bright region.



The diffraction of light by a circular opening



$$\sin\theta = 1.22\frac{\lambda}{D},\ \lambda = \mathit{wavelength},\ D = \mathit{Diameter}$$